Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H\times W\times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

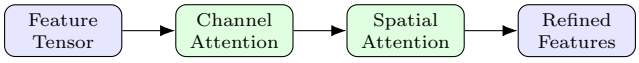
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

Which spatial features are related to other spatial features?

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time
3×3			
5×5			
7×7			

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.
Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.
To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention						
Channel / SE						
Spatial Attention						
ECA						
CBAM						
Self-Attention						
Additional Mechanism						

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.

For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.

The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	**Position**	**Accuracy**	**Precision**	**Recall**	**F1**	**Parameters**	**Inference Time**
Baseline	–						
Channel Attention	P_3						
Spatial Attention	P_3						
Channel+Spatial	P_2						
Channel+Spatial	P_3						
Channel+Spatial	P_4						
Self-Attention	P_3						
Additional Attention-1	P_3						
Additional Attention-2	P_3						

24. Graphical Performance Analysis

Students should not present only numerical tables.

Appropriate graphs must be generated to analyze model behavior.

All six graphical comparisons in Sections 24.1–24.6 are mandatory and must be included in the final report. Use the same test set and report measured values; do not invent or estimate results.

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.

Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.

Students should compare the convergence behavior of the baseline and attention models.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

$$\text{Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.}$$

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score}.$$

This comparison is useful when the class-wise performance is not uniform.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy}.$$

Each point should represent one attention mechanism.

This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad}-\text{CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input | Baseline Grad-CAM | Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

$$\boxed{\text{Input} \mid \text{Spatial Attention Map} \mid \text{Attention Overlay}}$$

Students should identify the image regions receiving high attention values.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

$$\boxed{\text{3 correctly classified images}}$$

and

$$\boxed{\text{2 incorrectly classified images.}}$$

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

31.1 Mandatory Figure Slots

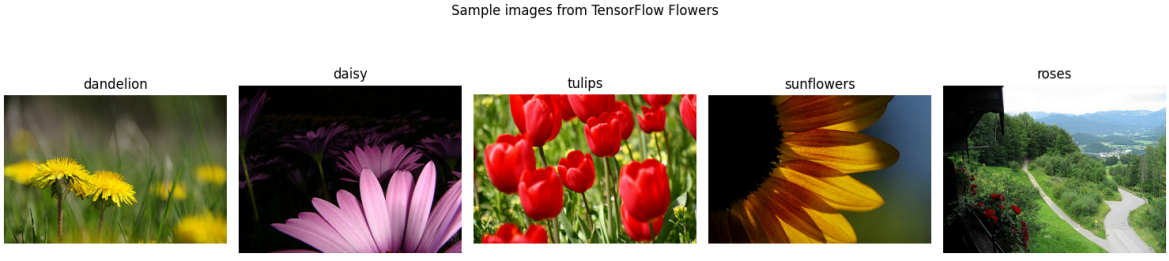


Figure 1: Representative input images from all five flower classes.

Inference: The images shows different types of flowers with variations in colour, shape and background. This helps the model learn different flower features and classify them into the correct classes.

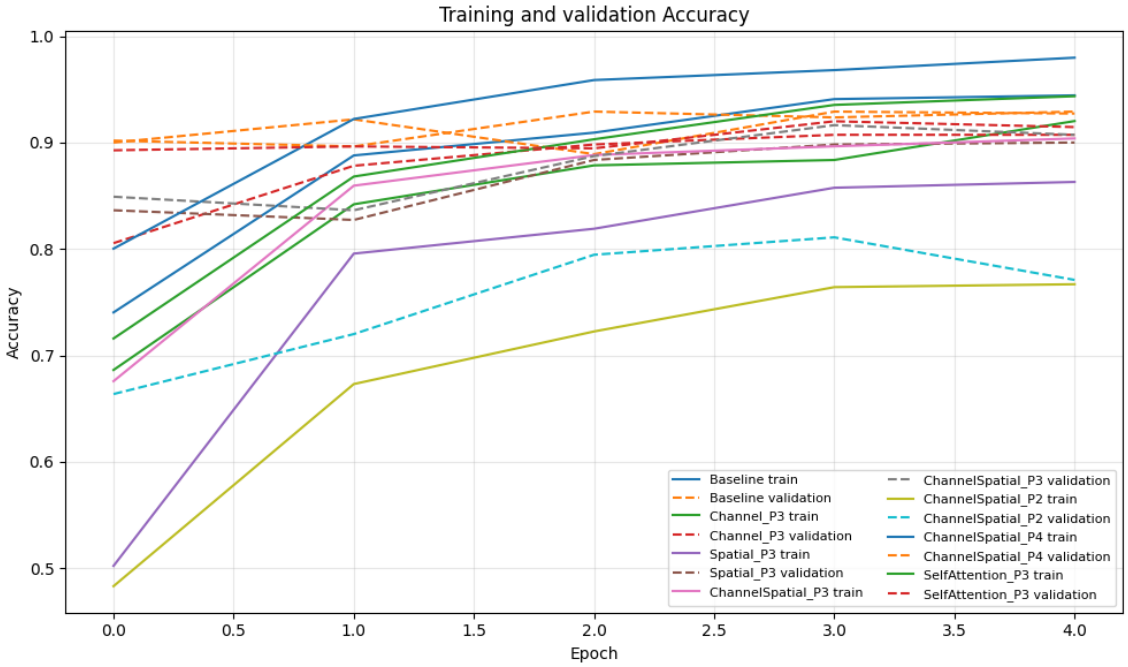


Figure 2: Training and validation accuracy versus epoch for the baseline and selected attention models.

Inference: The accuracy generally increases as the number of epochs increases, showing that the models are learning from the images. However, some models have a difference between training and validation accuracy, which may indicate slight overfitting.

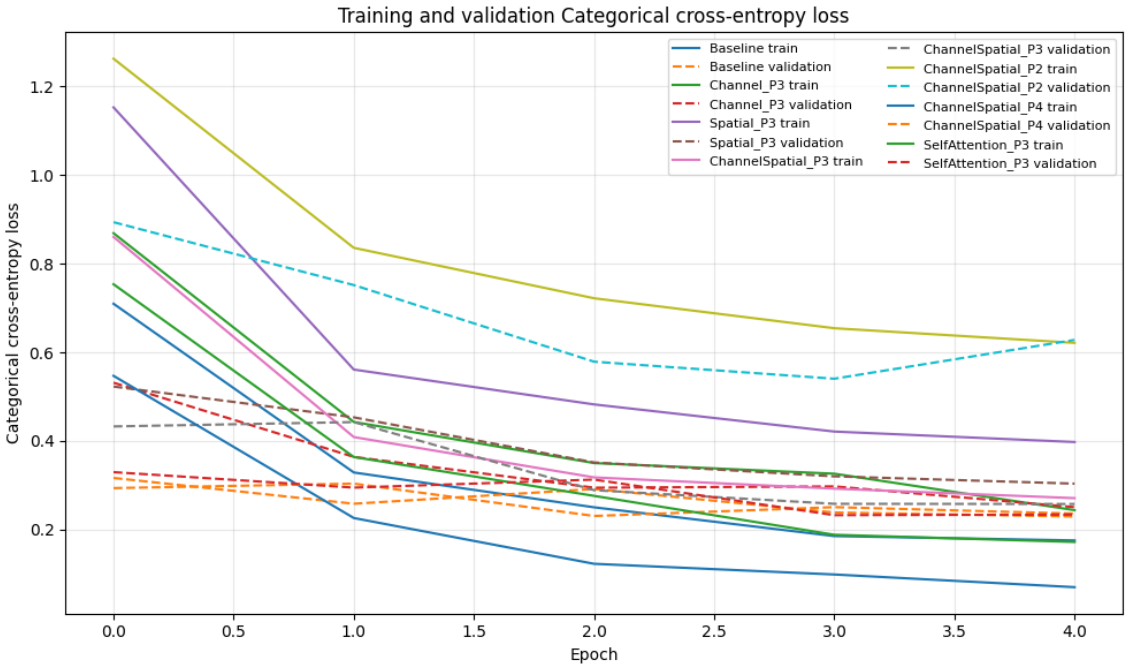


Figure 3: Training and validation loss versus epoch for the baseline and selected attention models.

Inference: The loss generally decreases during training, which indicates that the models are improving their predictions. Some curves become almost stable in later epochs, showing that the learning process is slowing down.

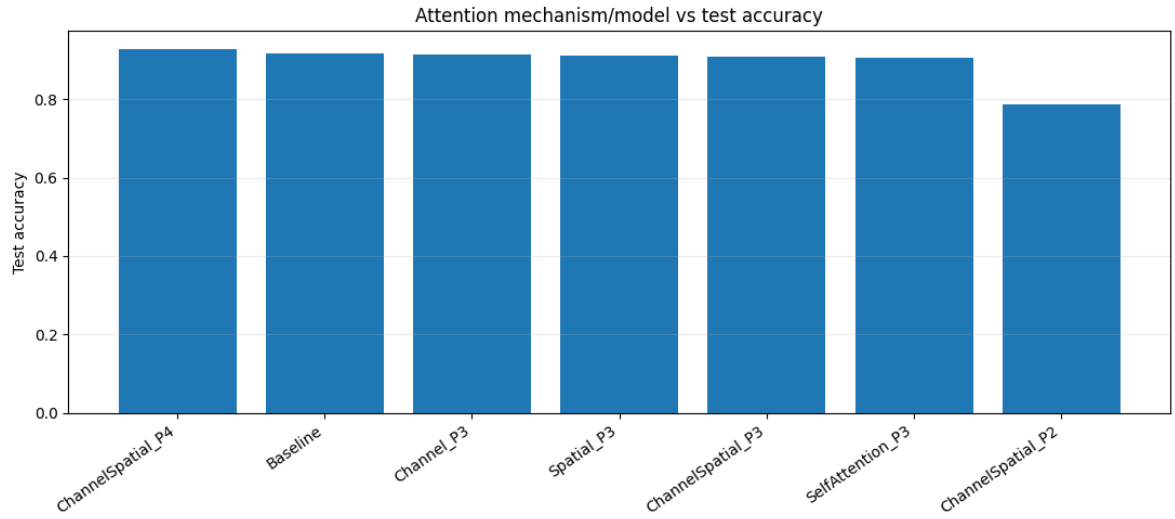


Figure 4: Test accuracy comparison across the baseline and attention mechanisms.

Inference: Most attention models shows similar accuracy, while one of the models has a comparatively lower score. This indicates that adding attention does not always improve classification accuracy.

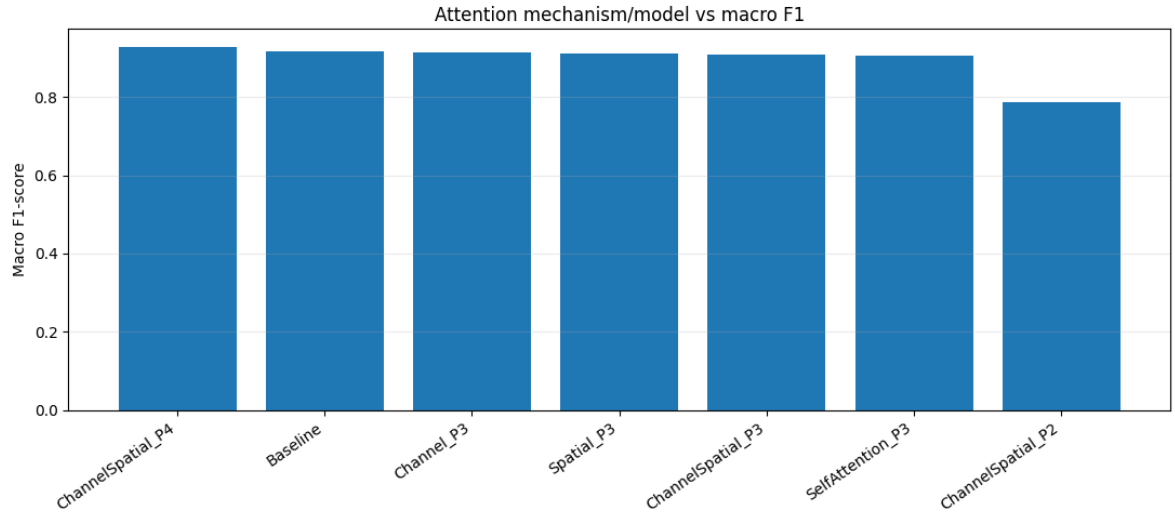


Figure 5: Test macro F1-score comparison across attention mechanisms.

Inference: The macro F1-scores are high for most of the models, which suggests a good balance between precision and recall across the flower classes. The lower score of one model shows that different attention mechanisms can affect class-wise performance.

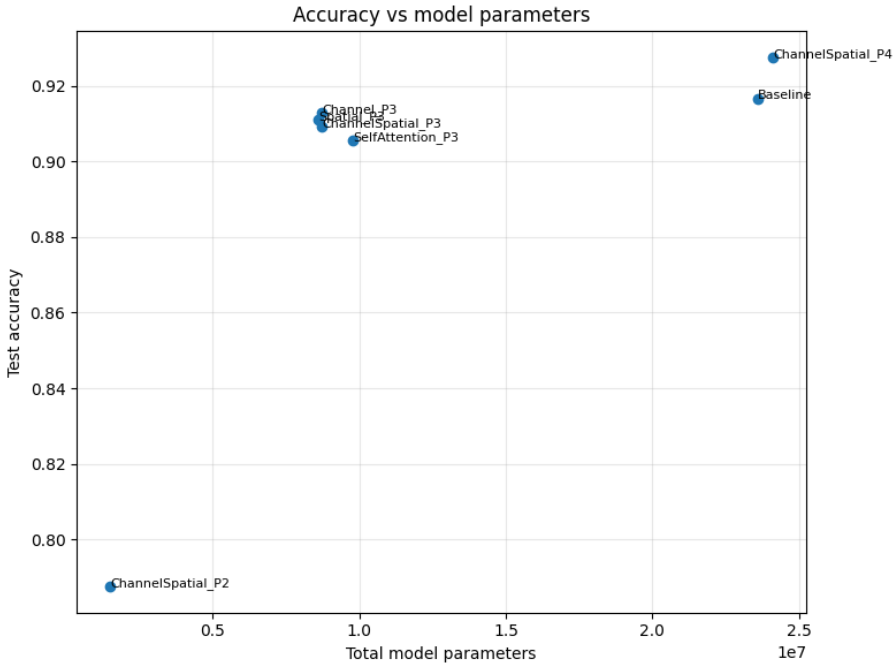


Figure 6: Test accuracy versus total model parameters.

Inference: The graph shows that models with different numbers of parameters can achieve different accuracy levels. A model with more parameters does not always perform better, so both accuracy and model complexity should be considered.

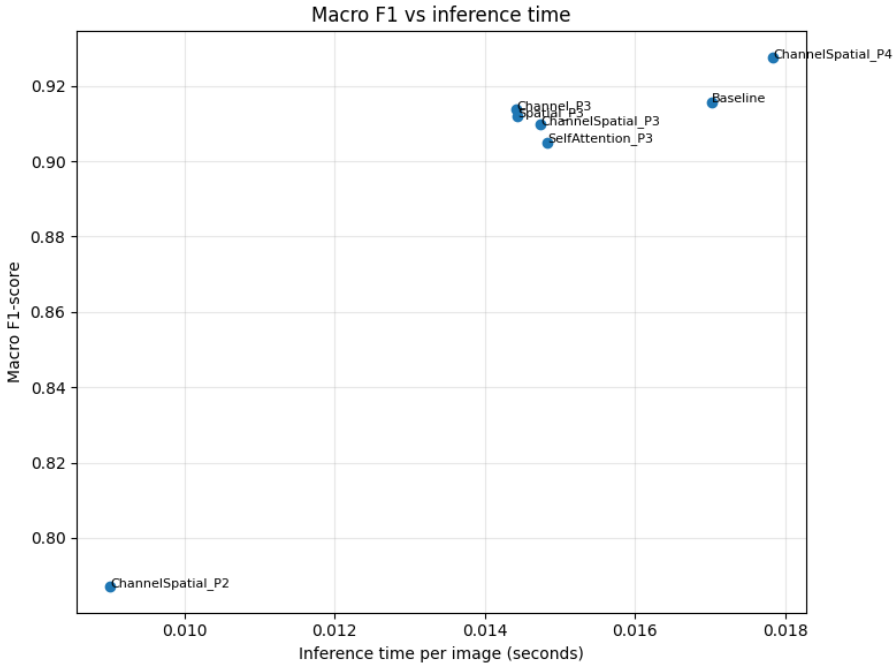


Figure 7: Test macro F1-score versus average inference time per image.

Inference: The graph compares the classification performance with the time taken to predict an image. Some models achieve a high F1-score with less inference time, which can make them a more practical choice.

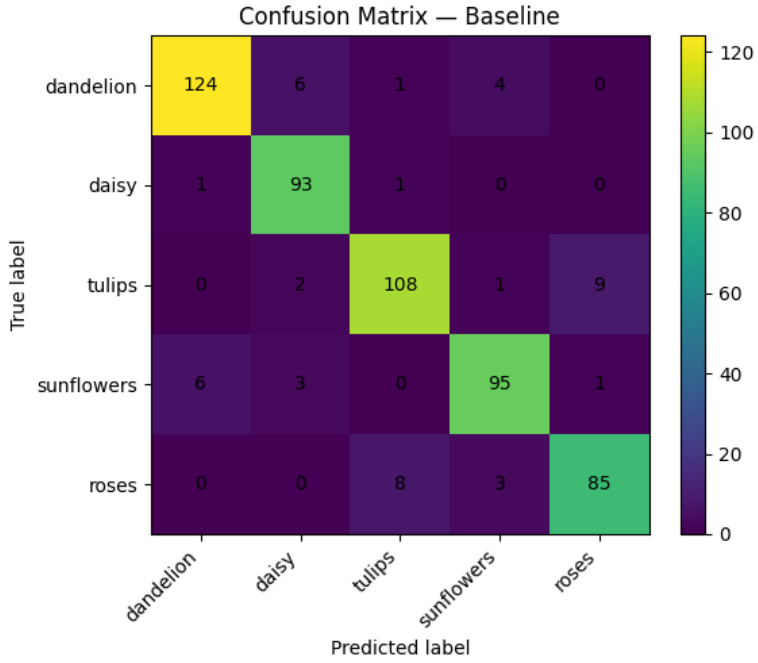


Figure 8: Confusion matrix for baseline ResNet50.

Inference: The confusion matrix shows that the baseline model correctly classifies many of the flower images, as seen from the values along the diagonal. Some images are still misclassified into other classes, showing that certain flowers are difficult to distinguish.

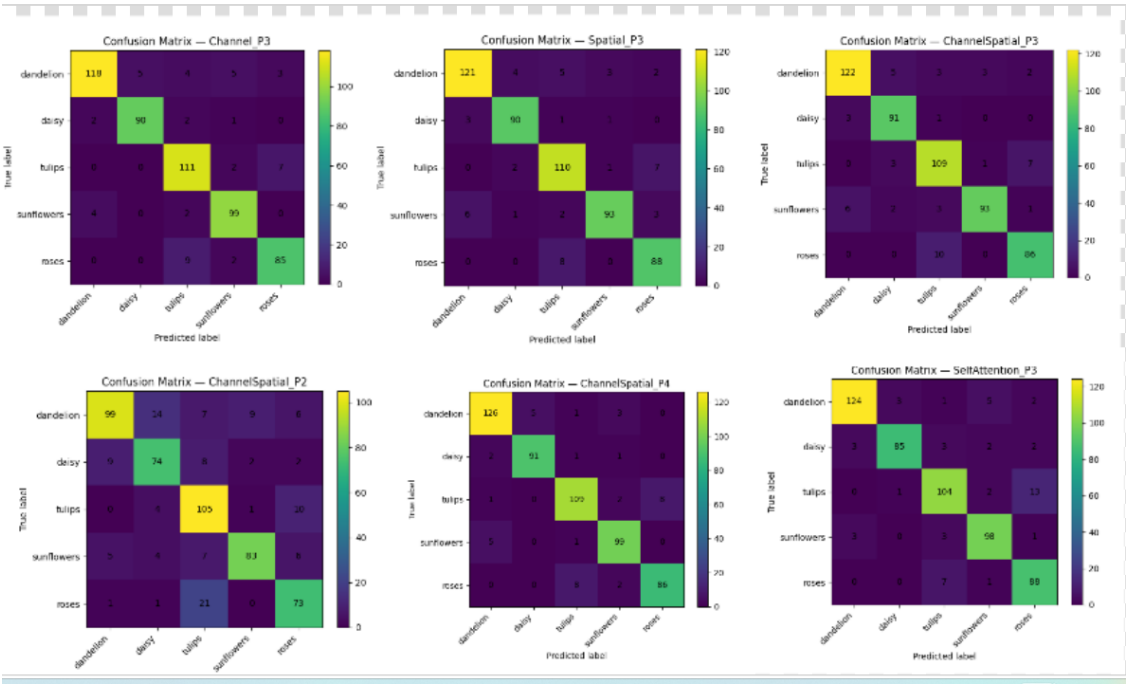


Figure 9: Confusion matrix for the selected attention-based ResNet50.

Inference: The confusion matrices help us compare the classification results of different attention-based models. The diagonal values represent correct predictions, while the other values show misclassified images, helping identify which flower classes are confusing the model.

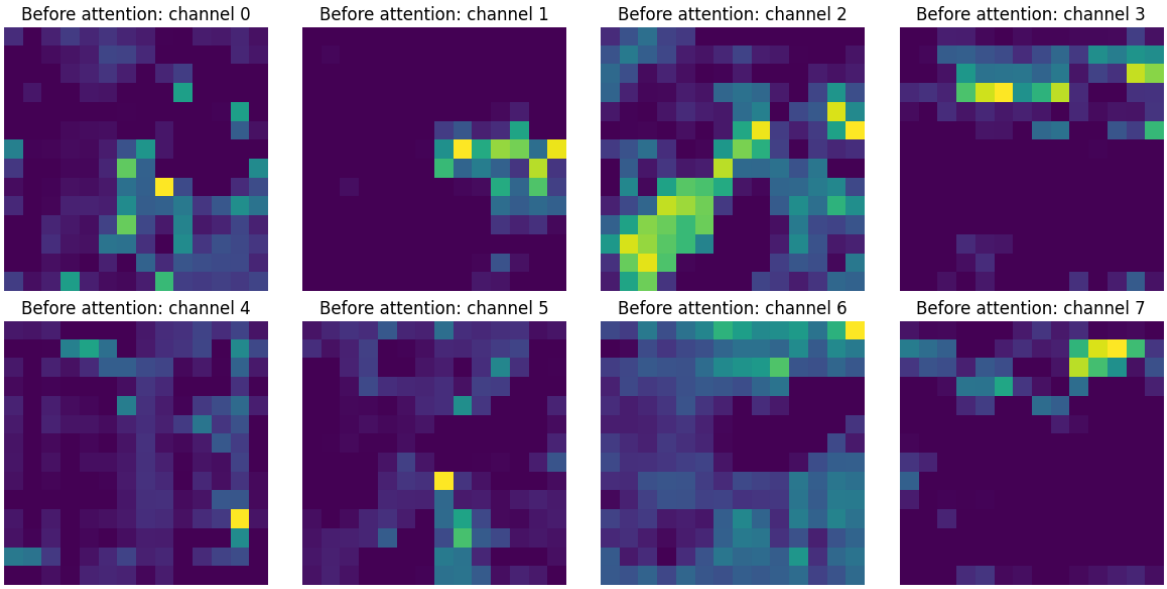


Figure 10: Example feature-map visualization immediately before the attention module.

Inference: The feature maps show that ResNet50 extracts different patterns from the input image before applying attention. Some maps highlight edges and textures, while others capture more meaningful structures that can help in flower classification.

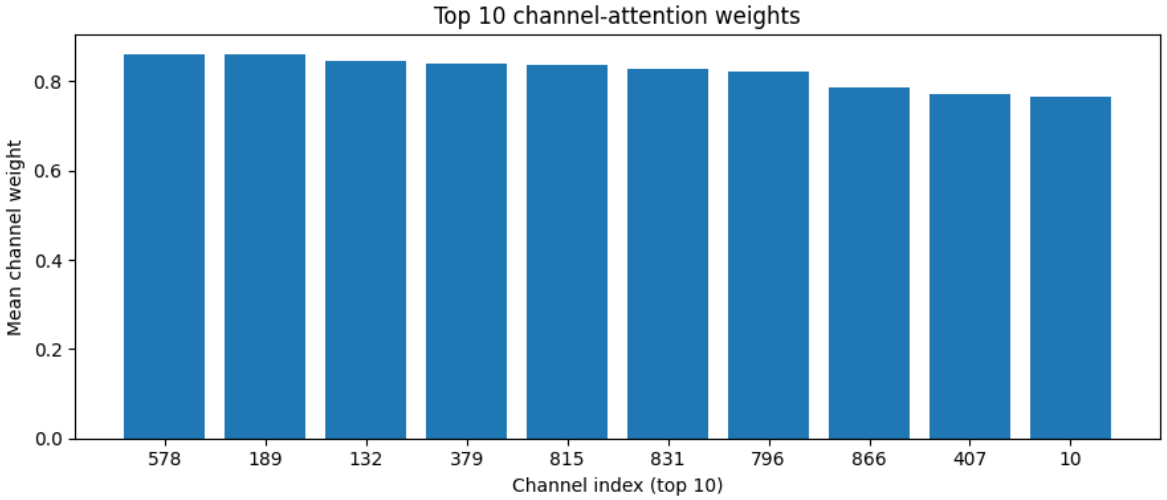


Figure 11: Top channel-attention weights for a representative test image.

Inference: The bar graph shows the importance assigned to the selected feature channels by the channel-attention mechanism. Some channels receive slightly higher weights than others, meaning the model gives more importance to certain features during classification

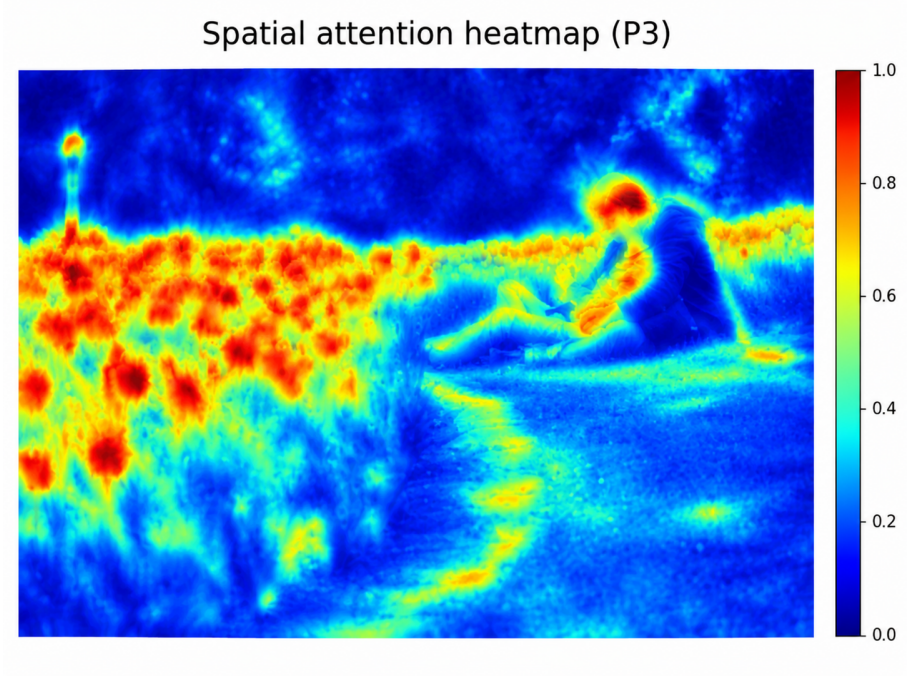


Figure 12: Spatial-attention heatmap for a representative test image.

Inference: The heatmap shows the spatial regions receiving higher attention values, represented by warmer colours. This helps us understand which parts of the feature map the model considers more important, although not all highlighted regions may be relevant to the flower.

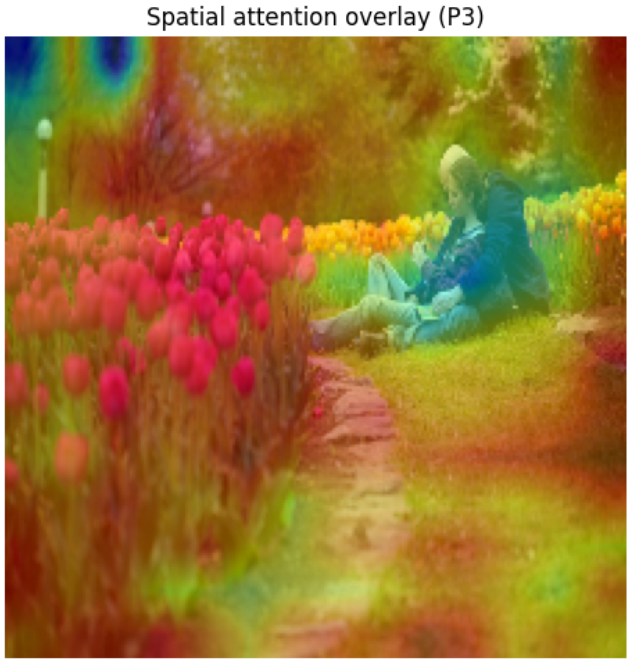


Figure 13: Spatial-attention heatmap overlaid on the input image.

Inference: The overlay combines the attention heatmap with the original image, making it easier to identify the important regions. The highlighted areas show where the model is focusing more, but some background regions also receive attention.



Figure 14: Spatial self-attention response for the self-attention experiment.

Inference: The self-attention map shows how different spatial positions in the feature map respond to one another. Some regions have stronger responses than others, indicating that the model learns relationships between different parts of the image.

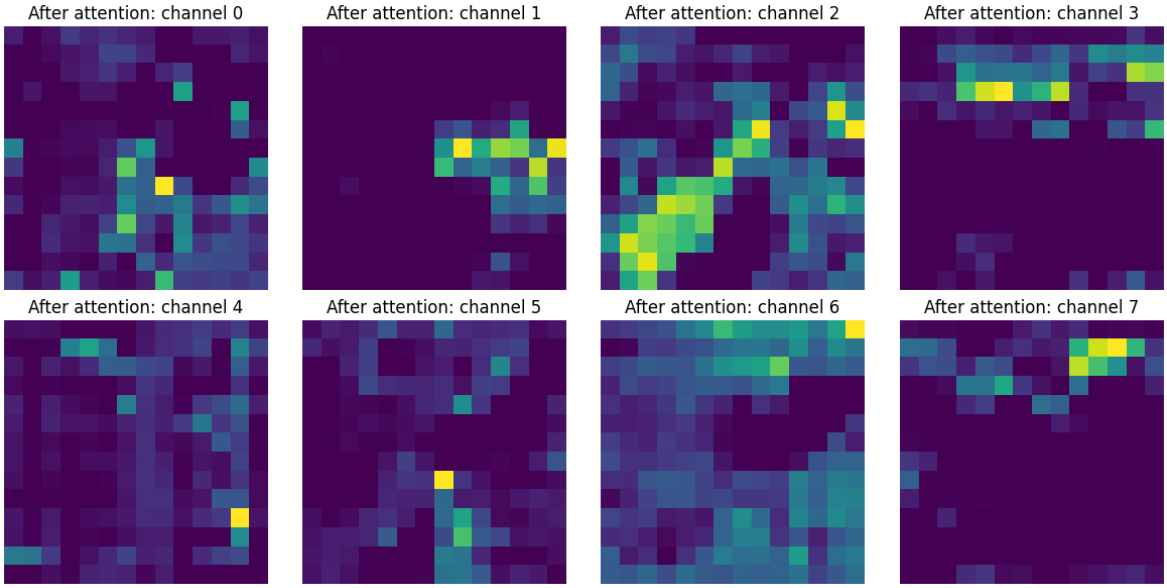


Figure 15: Example feature-map visualization immediately after the attention module.

Inference: The feature maps after attention show changes in the feature activations compared to those before attention. Some patterns become more prominent while others are reduced, showing how attention modifies the information used for classification.

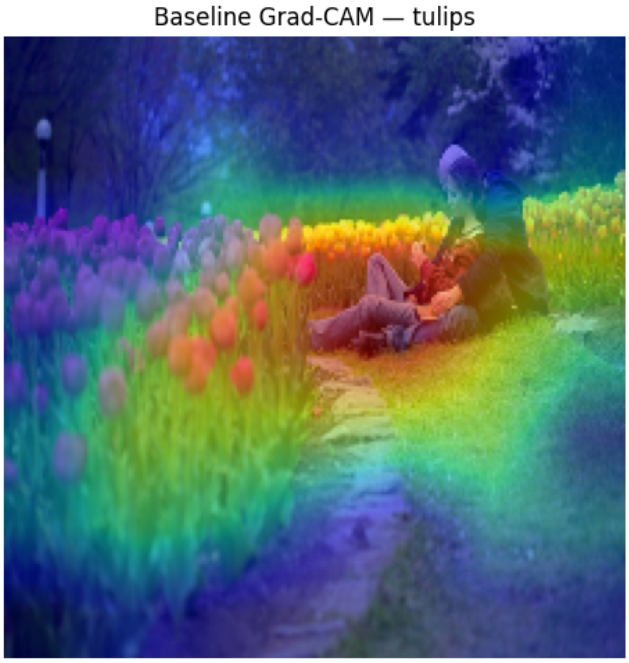


Figure 16: Baseline ResNet50 Grad-CAM for a selected test image.

Inference: The Grad-CAM visualization highlights the regions that contributed to the baseline model's prediction of tulips. The activation is spread across the flower and nearby areas, suggesting that the model uses visual information from more than just the flower petals.

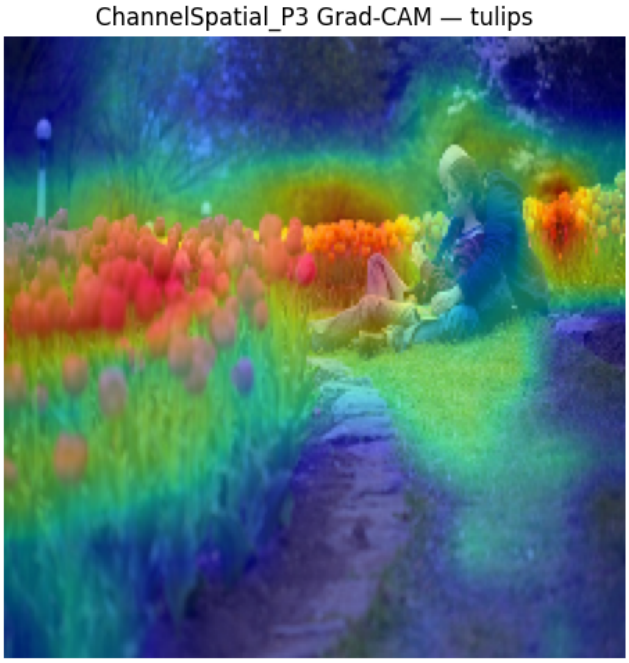


Figure 17: Attention-model Grad-CAM for the same test image.

Inference: The attention-based model shows a different activation pattern for the same test image. The highlighted regions appear more concentrated around some flower structures, although background areas are still visible in the heatmap.

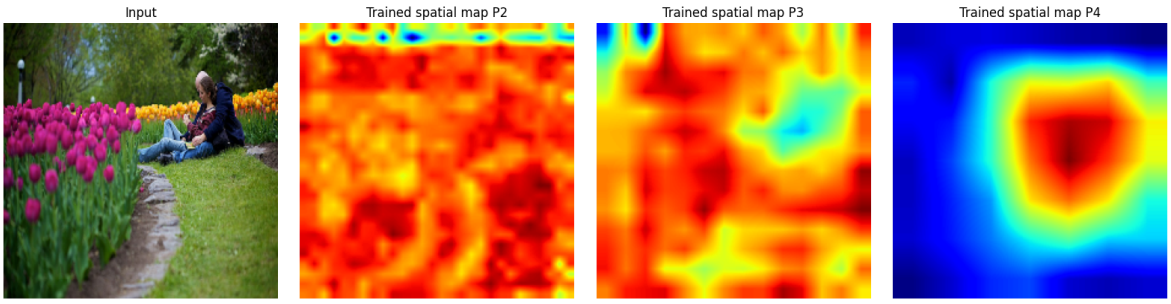


Figure 18: Spatial-attention maps at P_2, P_3, and P_4 for the same input image.

Inference: The attention maps differ depending on where attention is added in ResNet50. Earlier stages can preserve finer image details, while deeper stages may focus on broader flower regions, showing that attention position affects the features being highlighted.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

Parameters + Inference Time

33.3 Qualitative Analysis

Report:

Grad-CAM + Attention Maps + Visual Interpretation

Therefore, the overall evaluation should be:

Quantitative + Computational + Qualitative

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

$$\boxed{\text{Attention Position}}.$$

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

$$\boxed{\text{SE,} \quad \text{ECA,} \quad \text{CBAM,} \quad \text{Coordinate Attention,} \quad \text{Self-Attention}}$$

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;

- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?**
   Residual connections allow information to skip some layers, making deep networks easier to train. They also help reduce the vanishing gradient problem.
2. **What does conv2_x represent compared with conv5_x?**
   conv2_x learns basic features like edges and textures, while conv5_x learns complex features that help identify the flower class.
3. **Why does the spatial resolution decrease with network depth?**
   The network reduces spatial resolution using strided convolutions and pooling. This reduces computation and helps learn more meaningful features.
4. **Why does the number of feature channels increase with network depth?**
   Deeper layers need more channels to represent complex patterns and features. This helps the model understand the image better.
5. **What is meant by low-level, intermediate, and high-level features?**
   Low-level features include edges and colors, intermediate features include shapes and textures, and high-level features represent meaningful objects or flower characteristics.
6. **What is the purpose of an attention mechanism?**
   Attention helps the model focus on important information and reduce the influence of less useful features, which may improve classification performance.
7. **What does channel attention attempt to learn?**
   Channel attention learns which feature channels are more useful. It gives higher weights to important channels and lower weights to less useful ones.
8. **What does spatial attention attempt to learn?**
   Spatial attention identifies important locations in a feature map. For example, it may focus more on flower petals than the background.
9. **How is channel attention different from spatial attention?**
   Channel attention focuses on *what features* are important, while spatial attention focuses on *where* those features are located in the image.
10. **Why can channel and spatial attention be used sequentially?**
    They focus on different information, so they can work together. Channel attention selects useful features, while spatial attention highlights important locations.
11. **Why might channel attention behave differently at conv3_x and conv5_x?**
    conv3_x contains more textures and simple patterns, while conv5_x contains complex semantic features. So, channel attention may select different features at each stage.
12. **Why can spatial attention be useful at an intermediate ResNet stage?**
    Intermediate stages preserve useful spatial details while learning meaningful patterns. Spatial attention can highlight important flower regions before more details are lost.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?**
    Self-attention compares spatial positions with one another. More positions require more computations and memory, making it expensive at high resolutions.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?**
    The attention matrix has size $(HW) \times (HW)$, where H and W are the feature map dimensions. Increasing both dimensions makes the matrix grow quadratically with the number of positions.
15. **How does attention performance vary when the insertion position is changed?**
    Performance depends on where attention is added in ResNet50. Some stages may improve accuracy, while others may provide little benefit or even reduce performance.
16. **How does attention at P_2 differ visually from attention at P_4?**
    P_2 usually highlights fine details like edges and petal textures, while P_4 often shows coarser, more meaningful regions of the flower.

17. **Does applying attention at multiple stages always improve classification performance?**
    No, adding attention at every stage does not always improve accuracy. It can increase computation and complexity without providing any useful performance gain.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**
    Yes, the order changes how features are refined. Channel→Spatial selects useful channels first, while Spatial→Channel highlights important locations first.
19. **How does the spatial-attention kernel size influence the attention map?**
    A smaller kernel focuses on nearby details, while a larger kernel captures a wider spatial context. However, a larger kernel does not always give better results.
20. **How does SE attention differ from ECA attention?**
    SE uses fully connected layers to learn channel weights, while ECA uses a lightweight one-dimensional convolution. ECA generally adds fewer parameters and less complexity.
21. **How does CBAM combine channel and spatial attention?**
    CBAM first applies channel attention and then spatial attention. This helps the model identify both useful features and important image locations.
22. **How does self-attention differ from channel attention?**
    Channel attention learns which channels are important, while self-attention learns relationships between spatial positions. Self-attention can capture relationships between distant image regions.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**
    No, more parameters do not always mean better accuracy. A simpler attention mechanism may perform better with less computation and a lower risk of overfitting.
24. **What is the difference between an internal attention map and Grad-CAM?**
    An internal attention map helps refine features inside the model, while Grad-CAM creates a heatmap to explain which regions influenced a particular prediction.
25. **Why is Grad-CAM called a post-hoc explanation technique?**
    Grad-CAM is applied after model training to explain its predictions. It uses activations and gradients without changing the trained model.
26. **Does the attention model suppress irrelevant background regions?**
    It may reduce the importance of background regions and focus on the flower. However, attention does not always remove irrelevant information completely.
27. **Are the attention maps of correctly classified and incorrectly classified images different?**
    They can be different, as incorrect predictions may focus on confusing or irrelevant regions. However, this depends on the image and the trained model.
28. **Which flower classes benefit most from attention?**
    Visually similar flower classes may benefit because attention can highlight small differences in petals, colors, and textures. The actual classes should be identified from experimental results.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?**
    A confusion matrix shows which classes are correctly classified and which are often confused with others. Overall accuracy alone does not provide this class-wise information.
30. **Why is macro F1 useful in addition to accuracy?**
    Macro F1 gives equal importance to every class by averaging their individual F1-scores. It helps identify poor class-wise performance even when overall accuracy is high.
31. **What is meant by the accuracy–complexity trade-off?**
    It means balancing classification accuracy with model complexity. A more complex model may improve accuracy but require more memory and computation.
32. **What is meant by the F1–inference-time trade-off?**
    It describes the balance between classification quality and prediction speed. A model with better F1 may take longer to make predictions.
33. **Why must every ablation study use the same dataset split?**
    Using the same dataset split ensures a fair comparison between models. Otherwise, differences in performance may come from the data rather than the attention mechanism.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?**
    The same learning rate, optimizer, epochs, and other settings should be used for a fair comparison. This helps determine whether performance changes are actually due to attention.
35. **What conclusions can be drawn from the Grad-CAM comparison?**
    Grad-CAM shows which image regions influence the model's predictions. It can help identify whether the model focuses on flower features or background regions, but it does not prove the model's reasoning.

37.1 Discussion Questions: Model Answers

Important note: The answers below provide the theoretical discussion. Questions about which model performed best, which classes benefited, and what the plots show must be completed using the student's actual measured results. No accuracy, F1-score, timing, or visual finding should be invented.

1. **Purpose of residual connections:** Residual/skip connections add a block's input to its transformed output. They help gradients flow through deep networks and make optimization of deep models easier.
2. **conv2_x versus conv5_x:** conv2_x produces relatively high-resolution, lower-level features; conv5_x produces lower-resolution, higher-level semantic features.
3. **Why resolution decreases:** Strided convolutions and pooling progressively downsample the feature maps, reducing spatial computation while increasing receptive field.
4. **Why channels increase:** Deeper stages use more channels to represent a wider variety of increasingly complex patterns and semantic concepts.
5. **Feature levels:** Low-level features include edges, colors, and textures; intermediate features include parts and local patterns; high-level features encode larger object structures and class-relevant information.
6. **Purpose of attention:** Attention learns weights that emphasize useful features and reduce the influence of less useful features for the task.
7. **Channel attention:** It learns the relative importance of feature channels—roughly, which kinds of feature responses should be emphasized.
8. **Spatial attention:** It learns a spatial weight map that emphasizes selected locations in a feature map.
9. **Difference:** Channel attention weights channels and is typically shared across spatial locations; spatial attention weights locations and is typically shared across channels.
10. **Sequential channel–spatial attention:** The channel module first recalibrates feature types; the spatial module then selects locations using those recalibrated features. The order may change the result.
11. **Channel attention at conv3_x versus conv5_x:** The stages contain different channel counts, resolutions, and feature semantics. Therefore, the descriptor and learned channel weights operate on different representations.
12. **Why intermediate spatial attention can help:** Intermediate maps retain more spatial detail than late maps while containing more object-part information than early maps. This can help emphasize discriminative flower regions, although the experiment must verify it.
13. **Why self-attention is expensive at high resolution:** For $N = HW$ spatial positions, the pairwise attention matrix has N^2 entries, so memory and computation grow quadratically with the number of positions.
14. **Effect of increasing resolution on the attention matrix:** Doubling both height and width makes N four times larger and the $N \times N$ matrix sixteen times larger.
15. **Effect of insertion position on performance:** It must be determined from the position ablation. Compare the same channel–spatial module at P_2, P_3, and P_4, then identify the best measured metric and its computational cost.
16. **Visual difference between P_2 and P_4:** P_2 has finer spatial detail and may respond to edges, texture, and small parts; P_4 is coarser and may emphasize broader semantic regions. Describe only what the generated maps actually show.
17. **Does multi-stage attention always help?** No. Repeated attention can help, have little effect, or hurt due to redundant reweighting, optimization changes, overfitting, and added computation.
18. **Channel→Spatial versus Spatial→Channel:** They are generally not equivalent because the second module receives features already modified by the first. Compare them at the same stage and under the same training protocol.
19. **Effect of spatial kernel size:** A larger kernel uses a wider local neighborhood and may capture broader context, but it adds computation and may blur fine spatial distinctions. The best size is empirical.
20. **SE versus ECA:** SE uses global pooling followed by a small bottleneck transformation to model channel dependencies. ECA uses a lightweight one-dimensional convolution over channel descriptors, typically avoiding dimensionality reduction.
21. **How CBAM combines attention:** CBAM applies channel attention and then spatial attention sequentially to refine the feature tensor.
22. **Self-attention versus channel attention:** Self-attention models relationships between spatial positions through query–key interactions; channel attention reweights feature channels and need not explicitly calculate all spatial-position pairs.
23. **Does the most parameter-heavy mechanism always win?** No. More parameters do not guarantee better generalization or accuracy. Evaluate test metrics, training stability, inference time, and parameter count together.
24. **Internal attention map versus Grad-CAM:** An internal attention map is produced by the model's attention module during its forward pass. Grad-CAM uses gradients of a target class score with respect to convolutional feature maps to create a class-specific localization map.
25. **Why Grad-CAM is post-hoc:** It is computed after the model is trained or used for prediction as an explanation; it is not itself the learned attention operation that produces the prediction.
26. **Does attention suppress background?** This cannot be answered universally. Compare the spatial-attention

overlay and baseline/attention Grad-CAM on the same images and state whether the highlighted background visibly decreases.

27. **Correct versus incorrect predictions:** Maps may be more object-focused for correct predictions and may highlight background, leaves, or non-discriminative regions for errors, but this is an empirical observation, not a guarantee.
28. **Which classes benefit most?** Compare per-class recall/F1 or the confusion matrices before and after attention. Name the class with the largest measured improvement; do not infer it from overall accuracy alone.
29. **What the confusion matrix reveals:** It shows true-versus-predicted class counts, exposing specific class confusions and class-wise error patterns that a single overall accuracy value hides.
30. **Why macro F1 in addition to accuracy?** Macro F1 calculates F1 separately for each class and averages the scores equally, so each class contributes equally regardless of its sample count.
31. **Accuracy–complexity trade-off:** It is the balance between classification performance and the resources needed, such as parameter count, memory, and computation.
32. **F1–inference-time trade-off:** It describes whether a change in macro F1 justifies the change in prediction latency. Prefer a faster model when its performance is comparable and latency matters.
33. **Why use the same data split?** Different splits can have different difficulty and class composition. Keeping the split fixed makes comparisons more attributable to the model change.
34. **Why keep the training protocol fixed?** Changing the optimizer, epochs, augmentation, or learning rate alongside the attention module confounds the comparison; the effect of attention would no longer be isolated.
35. **Conclusions from Grad-CAM:** Grad-CAM can show which regions contribute to a class score and whether these regions differ between models. It does not prove causal reasoning or guarantee that a prediction is correct. Support conclusions with multiple examples and quantitative metrics.

Suggested result-based discussion paragraph

After filling in the results, use a paragraph in this format:

> Compared with the baseline, the *[attention model]* achieved a test accuracy of *[value]* and macro F1-score of *[value]*. Its parameter count and mean inference time were *[values]*. The position ablation showed that *[position]* performed best under the selected metric. In the confusion matrices, *[class pair]* was the most frequent confusion for the baseline and *[describe change]* after attention. For the same representative images, the attention map/Grad-CAM *[describe visible region differences]*. These results suggest *[careful conclusion]*, although the conclusion is limited by *[number of runs, dataset size, or other limitation]*.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**

2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

42. References

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10. TensorFlow Datasets: https://www.tensorflow.org/datasets

GitHub Repository:

https://github.com/shaliniparvathareddy/DL_lab8-9/blob/main